Product Information
Acronal® EDGE 7685

Polymer Dispersion for long lasting low dirt pick-up premium exterior paint

Acronal® EDGE 7685 is an anionic dispersion. It consists of fine, well divided particles. It is a modified acrylic copolymer that exhibits excellent exterior durability and color retention, excellent Dirt pick-up resistance and Efflorescence resistance in comparison with straight/pure acrylic used in the premium segment. Acronal® EDGE 7685 is suited to manufacture paints over a broad range of applications and can be formulated at different gloss levels.

Chemical Nature:

Polymer Dispersion of a modified acrylic copolymer

Benefits

- Excellent dirt-pick-up resistance
- Excellent Color durability
- Excellent Efflorescence resistance

Features

- Excellent weatherability
- APEO free
- Good Opacity
- High Water Whitening Resistance
- High Gloss
- Tough & Flexible

Properties

Product specification*	Solids content	%	48 ± 1
	pH value	pH	7.5 – 9.0
	Viscosity at 23 °C, RV DV 3/20	mPa s	500 – 3000
Other properties of dispersions	Minimum film-forming temperature (ISO 2115)	°C	26
	Density (ISO 2811-1)	g/cm³	approx. 1.04
	Resistance to frost	ºC	> 0
	Type of dispersion		anionic

*The aforementioned data shall constitute the agreed contractual quality of the product at the time of passing of risk. The data are controlled at regular intervals as part of our quality assurance program. Neither these data nor the properties of product specimens shall imply any legally binding guarantee of certain properties or of fitness for a specific
purpose. No liability of ours can be derived therefrom.

Note
The information submitted in this publication is based on our current knowledge and experience. In view of the many factors that may affect processing and application, these data do not relieve processors of the responsibility of carrying out their own tests and experiments; neither do they imply any legally binding assurance of certain properties or of suitability for a specific purpose. It is the responsibility of those to whom we supply our products to ensure that any proprietary rights and existing laws and legislation are observed.

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Applications

Areas of application

Acronal® EDGE 7685 is used for long lasting exterior premium finishes.

Processing

It is advisable to disperse the pigments and extenders with wetting and dispersing agents such as Dispex CX 4320 and water-soluble polyphosphates in an alkaline medium in advance before the polymer dispersion is added. It is only when products with very high viscosity are being mixed in low-speed mixers that Acronal® EDGE 7685 should be added together with the wetting and dispersing agents.

Acronal® EDGE 7685 has very good compatibility with pigments and fillers.
Various thickeners such as cellulosic, HEUR, Polyether, HASE, ASE and Bentonite can be used to adjust the viscosity and workability of the paint. The choice of thickener depends on whether the coating is expected to be free-flowing or more thixotropic.

Coalescing agent needs to be added in order to enable the polymer to form a uniform film below ambient temperature. It is usually sufficient to add coalescent at a level of 3 - 6%, expressed as a proportion of the total dispersion. Short-chain alcohol and glycols improve the freeze-thaw resistance of paints, but they can not be used to lower the film-forming temperature. If possible, solvents should not be added direct to the polymer dispersion, they should be mixed with the pigment paste and then added.

Acronal® EDGE 7685, like all fine dispersions, has a tendency to foam. It is therefore necessary to add a commercial defoamer at the level of 0.3 – 0.6%.Trials should be carried out to test the effectiveness of the defoamer.

Acronal® EDGE 7685 is protected against attack by microorganisms, preservatives must still be added to the formulated products so as to ensure uniform quality even with prolonged storage. Compatibility and effectiveness must be determined by preliminary trials.

Customers have to carry out their own trials when developing and processing products based on Acronal® EDGE 7685. The compatibility of Acronal® EDGE 7685 with other ingredients of formulations, its effect on mixing process and its adhesion on different substrates etc., are affected by a variety of factors which are too numerous for us to take into account in our own trials. This includes testing its stability by storing it at ca.50℃ to confirm that its viscosity remains stable.

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Safety

General
The usual precautions for handling chemicals must be observed. These include the measures set out in the guidelines of the organizations responsible for safety at work, in particular, good ventilation and fume extraction at the workplace, care of the skin and the wearing of eye protection.

Safety Data Sheet
When using this product, the information and advice given in our **Safety Data Sheet** should be observed. Due attention should also be given to the **precautions** necessary for handling chemicals.

Labelling
According to all the data at our disposal, Acronal® EDGE 7685 does not need to be labelled as a dangerous substance or preparation as defined in the relevant local directives according to their current status.

Storage

Acronal® EDGE 7685 must not be allowed to come into contact during storage with metals or alloys that are susceptible to corrosion. During storage it is particularly important to ensure that containers are closed tightly; in storage tanks the air must always be saturated with water vapor. Undue heating must be avoided, as much exposure to frost.

Given adequate tank and storage hygiene Acronal® EDGE 7685 can be kept for about nine months at 10-30ºC.

To prevent problems with microorganisms we recommend post-stabilizing the product with biocides for storage.

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